DON BOSCO INSTITUTE OF TECHNOLOGY

Department of Electronics &Telecommunication Engineering

Review Questions for MSE – Electromagnetics & Antennas

1. State & explain the Maxwell's equations. (point form)
2. State & explain Gauss Law, Faradays Law, Ampere’s Law.
3. Derive continuity equation & explain its physical relevance.
4. Justify that magnetic flux lines do not diverge.
5. Explain Laplace and Poisson’s relations.
6. Derive an expression for the electric field intensity due to a conductor of infinite length and having charge density ρl
7. Write point form of Ampere's law and interpret the same.
8. Derive Maxwell’s equation in point form.
9. State and explain Maxwell’s equation in free space in integral and differential form. Hence, explain the difference between conduction and displacement current.
10. Derive an expression for magnetic field intensity due to finite long straight element.
11. Explain the significance of Poynting theorem.
12. State Poynting theorem. Write its final expression and explain the meaning of each term.
13. Explain the significance of Poynting theorem
14. State and explain Faraday’s Law in both the integral and differential form? Explain shortcomings of each of the form.
15. Explain the difference between conduction and displacement current with aid of Maxwell's Equations.
16. Four like charges of 40μC each are located at four corners of square. The square diagonal is 12m. Find the force on 200μC charge located 5m above the centre of square.
17. In free space, V=6xy2z+8. At point P (1,2,-5) find and volume charge density.
18. Write the characteristic of equipotential surfaces.
19. What are Gaussian surfaces?
20. What is the concept of work done? Explain potential.

21. Explain the concept of gradient of a potential function.

22. Explain conservative field property. Which Maxwell's equation best explain this property.